

SHREYAS GUPTA

+91 9151606122 | contactshreyasgupta@gmail.com | Chennai, India
LinkedIn | GitHub

PROFESSIONAL SUMMARY

AI Researcher and Machine Learning Engineer pursuing Integrated M.Tech in Artificial Intelligence. Expertise in generative AI, agentic AI systems, multi-agent architectures, RAG pipelines, transformer-based models, and graph AI. Proven ability to design novel AI algorithms and build production-grade AI systems. Winner of Agentic AI Hackathon 2025 (NielsenIQ & TCS). Passionate about neural architectures, knowledge representation, and autonomous AI systems.

EDUCATION

SRM Institute of Science and Technology

Chennai, India

Integrated M.Tech in Artificial Intelligence (CGPA: 8.29/10)

Aug 2022 – May 2027

- Relevant Coursework: Machine Learning, Deep Learning, Neural Network Architectures, Statistical Learning Theory, NLP, Computer Vision, Graph Neural Networks, Reinforcement Learning, AI Agents & Multi-Agent Systems

EXPERIENCE

AI/ML Engineer – Research Intern

May 2026 – July 2026

Revenue

Bangalore, India

- Designed and implemented RAG pipelines and agentic AI architectures with multi-agent orchestration using Python, FastAPI, and LangChain across two enterprise AI products.
- Developed novel prompt engineering strategies and attention-based retrieval mechanisms, improving generative AI model performance by 35% on domain-specific benchmarks.
- Architected autonomous AI agent systems with tool-use capabilities, chain-of-thought reasoning, and knowledge representation modules for enterprise workflows.
- Shipped full-stack AI applications, migrated inference backends to AWS (EC2, Lambda, S3), and Dockerized ML model deployments for scalable serving.

Cybersecurity Research Intern – AI & Graph Systems

Dec 2025 – March 2026

Indian Institute of Technology, Guwahati

Guwahati, India

- Conducted research on graph-based AI models for vulnerability detection, designing modular architecture with CLI, parser, validator, loader, query engine, and UI modules.
- Implemented LadyBugDB – a custom embedded graph database with novel graph traversal algorithms for function call graphs and control flow edges, applicable to Graph AI and knowledge representation.
- Applied causal reasoning techniques to identify vulnerability propagation paths in dependency graphs, contributing to explainable AI for security analysis.

Open Source Contributor – AI Engineer

Dec 2025 – Present

NgKore Foundation (5G Core)

Remote

- Built autonomous Python AI agents leveraging LLM-based reasoning and multi-agent collaboration for monitoring, anomaly detection, and config management in 5G core deployments.
- Designed agent architectures with planning, tool-use, and self-correction capabilities, demonstrating practical agentic and multi-agentic systems implementation.
- Contributed features and fixes via GitHub PRs, reviewed and merged by senior maintainers.

Business Strategy & Data Analytics Intern

May 2024 – July 2024

To-Let Globe

Lucknow, India

- Applied statistical analysis on 10,000+ rental listings using SQL and Python to identify pricing and demand trends; built Power BI dashboards for leadership decision-making.

RESEARCH PROJECTS

- BharatGreen AI** | *Python, FastAPI, Next.js, TypeScript, NVIDIA NIM*

Jan 2026 – March 2026
- Built AI sustainability platform integrating generative AI models (NVIDIA NIM) for energy forecasting, carbon analysis, and water optimization across Indian regions.
 - Implemented what-if analysis engine using causal inference and temporal modeling for carbon-aware scheduling, advancing AI for UN SDGs.
 - Developed multimodal data pipeline handling temporal, geospatial, and text data using transformer-based architectures.
- Light Pollution Sentinel** | *Python, Next.js, PostgreSQL, OpenAI, Docker*

July 2025 – Sept 2025
- Built autonomous AI agent pipeline detecting light-pollution hotspots from NASA satellite data across 742 districts using concept discovery and extraction.
 - Developed explainable AI module generating severity-based alerts with natural language explanations.
 - Shipped production system with real-time heatmaps, AI-generated insights, and automated alerting.

AWARDS & ACHIEVEMENTS

- Winner – Agentic AI Hackathon 2025** | NielsenIQ & TCS

Sep 2025
- Built ALPS – an autonomous multi-agent system with LLM-based reasoning and orchestration for enterprise AI applications.
- Member – Indian Society for Technical Education (ISTE)** | SRM Institute

Mar 2025 – Present

TECHNICAL SKILLS

- AI/ML Frameworks:** TensorFlow, PyTorch, LangChain, OpenAI API, Google Gemini API, NVIDIA NIM, Hugging Face Transformers, scikit-learn, NLTK, Streamlit
- AI Research Areas:** Generative AI (LLMs), Agentic AI, Multi-Agent Systems, RAG, Transformer Architectures, Attention Mechanisms, Graph AI, Explainable AI, Causal Reasoning, Knowledge Representation
- Languages:** Python, Java, C/C++, SQL (PostgreSQL), JavaScript, TypeScript
- Frameworks & Tools:** FastAPI, React, Next.js, Node.js, Flask
- Infrastructure:** AWS (EC2, Lambda, S3), Azure, Docker, Linux, Git, GitHub, CI/CD
- Research Tools:** Jupyter Notebook, LaTeX, Weights & Biases, VS Code, Anaconda

CERTIFICATIONS

- TATA GenAI Powered Data Analytics, Forage (Oct 2025)
- Deloitte Technology Job Simulation, Forage (Jun 2025)
- Competitive Programming – Certificate of Excellence, Coding Ninjas (Jun 2024)
- Programming in Java, NPTEL (Oct 2023)